Soumodeep Karmakar

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Education

B.E. Electrical Engineering 2024 – Present
Jadavpur University, Kolkata *Kolkata, India*

- **CGPA:** 7.71 / 10.0 (Up to 3rd Semester)
- **Key Coursework:** Signals & Systems, DSP, Control Systems, Circuit Theory, Analog & Digital Electronics, Microprocessors, C Programming

Senior School Certificate Examination (Class XII) 2023
*Scottish Public School, Katihar (CBSE) – **84%** (Best of 5)*

Secondary School Examination (Class X) 2021
*Manipal Public School, Katihar (CBSE) – **92.2%** (Aggregate)*

Experience

Software Team Member Aug 2025 – Mar 2026
Jadavpur University Mechatronics Club (JUMTC) *Kolkata, India*

- **International Rover Challenge (IRC) 2026:** Developed a real-time **computer vision pipeline** for field deployment on a rover, enabling automated detection of mechanical tools across varied terrain and lighting conditions.
- Collected and manually annotated a custom dataset of mechanical tools using **Roboflow**, ensuring class balance and annotation accuracy for model training.
- Built and maintained the **Base Station dashboard** for live telemetry monitoring, aggregating sensor data streams from the rover through an operator-facing interface.
- Collaborated with a cross-functional team of 20+ students on system integration between the onboard vision module and ground station software.

Skills

Languages: Python, C/C++, JavaScript
Frameworks & Libraries: PyTorch, TensorFlow (basic), OpenCV, MediaPipe, Streamlit, Librosa
AI & ML: Deep Learning, Computer Vision, Object Detection (YOLOv8), Image Segmentation (UNet++), SAR/Multispectral Imagery, Signal Processing
Developer Tools: Git, GitHub, Docker, Roboflow, Kaggle, VS Code

Projects

Flood Semantic Segmentation | *UNet++, ResNet-101, CBAM, PyTorch, SAR+Optical* |

- Built a multi-class flood segmentation system (No Flood / Flood / Water Body) for West Bengal using **SAR (HH, HV) + multispectral (Green, Red, NIR)** satellite imagery from EOS-4 and Resourcesat-2.
- Designed a custom **CBAM-based learnable adapter** to project N-band satellite input into 3 channels, enabling fine-tuning of a frozen ImageNet ResNet-101 encoder without losing spectral context.
- Applied **Focal + Dice loss** with class weights to handle severe flood-class imbalance; used CutMix, GaussNoise, and RandomRotate90 augmentations to mitigate data starvation (59 training images).
- Implemented a **Flood-Priority Override** post-processing step using softmax probability thresholds to resolve spectral ambiguity between flood water and permanent water bodies.
- Achieved **Flood IoU: 0.1981** on the Phase 2 leaderboard – Top 4 Honorary Team, Top 7 Overall at AISEHack 2026, IIIT Hyderabad (IBM Theme).

Interactive Signal Plotter | *Python, Streamlit, NumPy, Matplotlib* |

- Developed an interactive web application using **Streamlit** to dynamically evaluate, combine, and visualize mathematical singularity functions (e.g., unit step, impulse, ramp).
- Engineered a custom **Object-Oriented** `SingularityFunction` class and implemented a secure evaluation environment to process and plot complex algebraic expressions entered by the user.

Mechanical Tool Detection System | *YOLOv8, PyTorch, Roboflow, Ultralytics*

- Designed and trained a custom **YOLOv8n** object detector to identify mechanical tools in real-world rover field conditions, achieving **0.80 mAP@0.50** on a curated annotated dataset.
- Addressed class imbalances through **Mosaic data augmentation** and hyperparameter tuning; exported model for **edge deployment** on low-power hardware.

Audio Deepfake Detection | *PyTorch, ResNet18, XGBoost, Librosa*

- Developed a hybrid deepfake detection system achieving **99.7% accuracy** by combining ResNet18 on mel spectrograms (512-dim embeddings) with 86 handcrafted features (MFCCs, Delta MFCCs, Chroma) for an XGBoost classifier.

- Ran ablations across ResNet-only, XGBoost-only, and fusion pipelines – fusion consistently outperformed both branches, confirming complementary information capture.

Netra Vaani | *Python, MediaPipe, OpenCV* |

- Built a real-time assistive communication system for **ALS patients**, enabling communication via eye movements using gaze direction tracking and blink detection.
- Implemented **Eye Aspect Ratio (EAR)** method with intelligent pause mechanisms to prevent false triggers during natural eye closure.

Micrograd | *Python, NumPy*

- Engineered a scalar-valued **autograd engine from first principles**, implementing reverse-mode automatic differentiation over dynamically built DAGs and training MLPs using SGD.

Achievements

AISEHack 2026 – Top 4 Honorary Team (Top 7 Overall) Apr 2026
IIIT Hyderabad | IBM Theme: Flood Detection *Jadavpur University*

- Cleared Phase 1 and reached the finale in the IBM-sponsored flood segmentation theme, achieving **Flood IoU: 0.1981** on the Phase 2 leaderboard – ranked in the top 7 teams overall.

Medha-Manthan 2025 – 1st Prize Sep 2025
IEEE Kolkata Chapter *Jadavpur University*

- Developed an efficient and versatile communication board for physically disabled people and won the first prize.

Hackforge – 3rd Position Mar 2025
Srijan'25, CodeClub Jadavpur University *Jadavpur University*